

Evaluation Techniques



Short Questions with Answers

1. What are Jakob Nielsen's heuristics?

Answer:

Jakob Nielsen's heuristics are **10 usability principles** used to evaluate user interfaces and identify potential usability issues without requiring user testing.

2. How many heuristics are there in Nielsen's model?

Answer:

There are **10 usability heuristics** defined by Jakob Nielsen.

3. What is the purpose of using heuristics in UI design?

Answer:

The purpose is to **improve user experience** by identifying and solving usability problems during the design or evaluation phase of a product.

4. What is the heuristic 'Visibility of System Status'?

Answer:

It means the system should **keep users informed** about what's happening through appropriate feedback (e.g., progress bars, loading indicators).

5. What does 'Match Between System and Real World' mean?

Answer:

The interface should use **familiar terms, icons, and metaphors** that reflect the user's real-world experience (e.g., trash bin icon for deleting files).

6. What does 'User Control and Freedom' mean?

Answer:

Users should have the ability to **undo actions or cancel operations**, allowing them to correct mistakes easily.

7. What is meant by 'Consistency and Standards'?

Answer:

The design should **follow platform conventions** so users don't have to guess what different elements mean (e.g., using a gear icon for settings).

8. What is 'Error Prevention' in heuristics?

Answer:

The system should be designed to **minimize user errors** with clear instructions and confirmation steps (e.g., asking before deleting a file).

9. What does 'Recognition Rather Than Recall' mean?

Answer:

The system should make options **visible and easy to access** rather than relying on users' memory (e.g., search autofill suggestions).

10. What is 'Flexibility and Efficiency of Use'?

Answer:

The interface should **support both beginners and advanced users**, offering shortcuts and customisation for experienced users.

11. What does 'Aesthetic and Minimalist Design' mean?

Answer:

The interface should be **clean and clutter-free**, showing only relevant information to avoid overwhelming the user.

12. What is the heuristic 'Help Users Recognize, Diagnose, and Recover from Errors'?

Answer:

Error messages should be **clear and helpful**, guiding users to solve problems (e.g., "Incorrect password" instead of "Login failed").

13. What is meant by 'Help and Documentation'?

Answer:

There should be **accessible guides or support documents** for users who need help (e.g., FAQs, guided tutorials, tooltips).

14. What is the purpose of user participation in evaluation?

Answer:

User participation helps ensure that systems are tested in **real-world conditions**, reflecting **actual user behavior and needs**.

15. What are the two main styles of evaluation?

Answer:

1. **Laboratory Studies** – Controlled settings for precise observation.
 2. **Field Studies** – Real-world environments for natural behavior analysis.
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16. What are subjective evaluation techniques?

Answer:

Methods based on **user opinions or experiences**, like **interviews, surveys, or heuristic evaluations**.

17. What are objective evaluation techniques?

Answer:

Methods based on **measurable, numeric data**, like **task completion rates, error rates, or time-on-task**.

18. What is Think-Aloud Protocol?

Answer:

A method where **users verbalize their thoughts** while performing tasks to uncover usability issues.

19. What is Heuristic Evaluation?

Answer:

An expert-based method where **usability professionals assess the interface** based on predefined principles.

20. What is A/B Testing?

Answer:

A method where **two versions** of a system are compared to determine which performs better.

21. What are low-level and high-level information in evaluation?

Answer:

- **Low-level:** Focuses on small UI details (e.g., button position).
 - **High-level:** Focuses on overall usability or workflow (e.g., task efficiency).
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22. What is immediate response evaluation?

Answer:

Evaluation requiring **real-time feedback**, such as usability testing or heatmaps.

23. What is high interference in evaluation?

Answer:

When the evaluator **actively guides or interacts** with users during the test, influencing their behavior.

24. What resources are needed for evaluations?

Answer:

- Equipment (labs, software)
 - Time
 - Money
 - Participants
 - Expertise
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Descriptive Questions with Answers

1. Explain the importance of Jakob Nielsen's heuristics in UI evaluation.

Answer:

Nielsen's heuristics provide a set of guidelines that help designers and developers **evaluate usability**, **identify pain points**, and **enhance user experience**. These principles are especially useful when conducting **expert reviews** of an interface without involving users directly. They cover key areas like **error handling**, **user control**, **consistency**, and **visual design**.

2. Describe any five of Nielsen's usability heuristics with examples.

Answer:

1. **Visibility of System Status**
 - Keep users informed about system operations with appropriate feedback.
 - *Example:* A progress bar when uploading a file.
 2. **User Control and Freedom**
 - Allow users to undo or cancel actions easily.
 - *Example:* An "Undo" button in a text editor.
 3. **Consistency and Standards**
 - Follow platform conventions and standards.
 - *Example:* Use a gear icon for settings across all platforms.
 4. **Recognition Rather Than Recall**
 - Minimize memory load by showing options.
 - *Example:* Autofill suggestions in search bars.
 5. **Aesthetic and Minimalist Design**
 - Remove unnecessary elements and keep the design clean.
 - *Example:* A dashboard with only essential metrics displayed.
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3. Why is 'Recognition Rather Than Recall' important in user interface design?

Answer:

Because users should not have to **memorize steps or commands**. Showing relevant options or suggestions makes the interface **easier to use, faster to navigate, and reduces user errors**.

4. How does 'Help Users Recover from Errors' improve usability?

Answer:

By providing **clear and meaningful error messages**, users understand what went wrong and how to fix it, reducing frustration and improving confidence in using the system.

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4. Explain the difference between laboratory and field studies.

Answer:

- **Laboratory Studies** are done in **controlled environments**, using tools like eye tracking or screen recordings to observe user interactions precisely.
 - **Field Studies** occur in **natural settings**, such as a user's workplace, offering more authentic insights into real-life behavior.
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5. What are the types of experimental design used in evaluation?

Answer:

- **Between-Subjects Design:** Different users test different versions.
- **Within-Subjects Design:** The same users test multiple versions.

These help compare system effectiveness based on performance metrics and user experience.

6. Describe at least three observational techniques.

Answer:

- **Think-Aloud Protocol:** Users speak their thoughts aloud while working.

- **Cooperative Evaluation:** Users and evaluators discuss issues during the task.
- **Post-Task Walkthrough:** Users review their actions with the evaluator after completing tasks.

7. Compare subjective and objective evaluation techniques.

Aspect	Subjective Techniques	Objective Techniques
Definition	Based on user opinions and feelings	Based on measurable performance data
Examples	Surveys, interviews, focus groups	Error rates, completion time, success rate
Data Type	Descriptive (qualitative)	Numeric (quantitative)
Best For	Understanding emotions and preferences	Comparing system performance

8. What factors influence the choice of evaluation method?

Answer:

- **Development Stage** (Design or Implementation)
- **Type of information needed** (Low vs. High-level)
- **Immediacy of response** (Real-time vs. Delayed)
- **Level of interference** (High vs. Low)
- **Resources available** (Time, tools, participants, money, expertise)

9. Why is user participation crucial in evaluation?

Answer:

Because it:

- Identifies real **usability issues**
- Reduces design flaws early
- Increases **user satisfaction and adoption**
- Provides both **objective data** and **subjective insights**